Marvin H. Cheng, Ph.D.

Coordinator, Center for Occupational Robotics Research
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Executive Profile

Engineering leader with over 20 years of experience leading national-scale research programs in robotics, cyber-physical systems, and safety-critical engineering across academia, federal agencies, and industry. Demonstrated success in managing multi-million-dollar portfolios, building multi-institutional partnerships, and advancing interdisciplinary research and workforce initiatives.

Currently leads an $8M+ national research center at NIOSH focused on robotics and workforce safety, coordinating multi-institutional research and federal initiatives. Serves in national leadership roles shaping robotics research and safety standards through NITRD IRAS and ANSI/A3.

Award-winning educator with demonstrated impact in mentoring and workforce development, with a focus on experiential, interdisciplinary engineering education. Positioned to advance academic leadership in research growth, industry engagement, and program development.

Strategic Leadership and Impact

- Lead a national robotics research center with an $8M+ multi-year portfolio, advancing workforce safety and autonomous systems
- Directed 30+ funded projects totaling $2.84M+, spanning federal, industry, and international sponsors
- Built and sustained multi-institutional collaborations with leading universities and federal agencies
- Oversaw six research laboratories and multiple doctoral-level research teams in robotics and cyber-physical systems
- Contributed to U.S. national robotics safety standards (ANSI/A3) and federal interagency initiatives (NITRD IRAS)
- Mentored 600+ students and trainees, supporting workforce development in robotics, AI, and safety-critical systems

Collectively, these efforts have advanced national capabilities in robotics safety, strengthened interdisciplinary research ecosystems, and contributed to workforce development pipelines in AI and autonomous systems.

Professional Experience and Leadership

National Institute for Occupational Safety and Health (NIOSH), CDC

- **Coordinator**, Center for Occupational Robotics | 2024 – Present
- **Assistant Coordinator** |2022 – 2024
  - Provide leadership for a nationally coordinated research program spanning academia, industry, and federal stakeholders with an $8M+ multi-year portfolio advancing robotics, automation, and workforce safety
  - Direct and expand multi-institutional collaborations with leading universities (UW–Madison, NCSU, Stony Brook, WVU, University of Florida, Carnegie Mellon)

- Provide strategic leadership for research initiatives aligned with the National Occupational Research Agenda (NORA)
- Serve as technical advisor to industry and federal agencies
- Lead interagency coordination through OSHA/NIOSH/A3 Alliance
- Establish CORR as a national platform for robotics safety research and workforce development

- **Team Lead**, Safety Controls Team (SCT) |2021 – present
- **Research General Engineer** | 2018 – present
  - Lead multiple doctoral-level research teams across robotics, control systems, and cyber-physical systems
  - Manage budgets and executions for multi-PI research portfolios, ensuring alignment with federal priorities and program goals
  - Oversee operation of six laboratories, supporting interdisciplinary research and workforce development

National Leadership and Standards

Co-Chair, Intelligent Robotics and Autonomous Systems (IRAS)
Networking and Information Technology Research and Development (NITRD) Program | 2024 – present

- Coordinate interagency robotics and autonomy research across federal agencies (NSF, NASA, DoD, NIST)
- Lead integration of research priorities and development of national-level reports and strategies

Voting Member, ANSI/A3 R15 Standard Committees | 2020 – present.

- Contribute to U.S. national robotics safety standards for industrial (ANSI/A3 R15.06) and mobile robotic systems (ANSI/A3 R15.08)
- Provide technical guidance on emerging technologies and safety best practices

Academic Appointments

Embry-Riddle Aeronautical University

- **Associate Professor**, Engineering | 2017 – 2018.
- **Adjunct Associate Professor** | 2018 – 2023.

West Virginia University

- **Assistant Professor**, Mechanical and Aerospace Engineering | 2010 – 2017.
- **Adjunct Professor** | 2018 – current.

Georgia Southern University

- **Assistant Professor**, Mechanical Engineering | 2006 – 2010

Education

Purdue University, West Lafayette, Indiana, December 2005
Ph.D. Mechanical Engineering

National Sun Yat-Sen University, Kaohsiung, Taiwan June 1996
Master of Mechanical Engineering

National Sun Yat-Sen University, Kaohsiung, Taiwan June 1994

Bachelor of Mechanical Engineering

Other Professional Experience

- **Technology, Innovation and Partnerships Program, National Science Foundation,** Program Director (selected, IPA), 06/2023**.**

Grants and Funded Research

Total Funding: $2.84M+ across federal agencies, industry, and international sponsors
Research Areas: Human–Robot Interaction, Occupational Safety, Robotics, Intelligent Systems, and Advanced Sensing

Federal Funding

- Centers for Disease Control and Prevention (CDC) / NIOSH (including NORA) – $1.38M+
  - Multiple awards as PI and Co-PI (2018–2026) supporting research in human–robot collaboration, workplace safety, mobile robotics, and safety technologies
  - Representative projects include:
    - Machine learning for worker safety in human-robot collaboration (PI)
    - Smart masonry robotics for struck-by hazard prevention (Co-PI)
    - Mobile robot safety, trust, and interaction in industrial environments
    - Driver-vehicle interface safety for law enforcement
    - Laboratory modernization and experimental infrastructure development
- National Aeronautics and Space Administration (NASA) – $857K
  - Centennial Sample Return Robot Challenge (Co-PI, multiple awards, 2014–2016)
  - Led development of advanced robotic systems for autonomous sample retrieval
- EPSCoR Programs. Total: $155K
  - PI on multiple projects advancing wearable robotics, synthetic sensing skin, and energy harvesting systems

Industry-Sponsored Research – $170K

International Government Funding – $243K

Publications, Scholarly Contributions, and Research Impact

Scholarly Output:

- 50+ peer-reviewed journal articles
- 40+ refereed conference proceedings
- 3 national industrial standards (ANSI/A3)
- Book chapters, technical reports, and federal publications

Research Areas:
Human–Robot Interaction, Robotics and Autonomous Systems, Cyber-Physical Systems, Occupational Safety and Health, Intelligent Control, Mechatronics, Sensors and MEMS, and AI-enabled engineering systems

Standards, Policy, and National Impact

- Contributor to U.S. national robotics safety standards (ANSI/A3 R15 series) for industrial and mobile robots
- Contributions to federal interagency initiatives (Intelligent Robotics and Autonomous Systems – IRAS Working Group)

- Publications and reports influencing national policy discussions on robotics safety and the future of work

Scholarly Impact Statement

My research advances human-centered robotics and safety-critical systems by integrating engineering innovation with public health and workforce protection. This work has contributed to both foundational research and applied solutions, influencing industry practices, federal initiatives, and interdisciplinary collaborations at the intersection of robotics, AI, and occupational safety.

Professional Service and Leadership

Demonstrated national and international leadership in human-centered robotics, occupational safety, and AI-enabled systems, with sustained contributions to federal initiatives, standards development, and global scholarly engagement.

National and International Leadership

- NSF Panel Reviewer and NSF Graduate Research Fellowship Reviewer (since 2010)
- Contributor to federal and interagency initiatives (NIOSH, NITRD IRAS)
- Voting Member, ANSI robotics safety standards committees
- Editorial roles and reviewer for IEEE, ASME, IFAC journals and conferences

Invited Keynotes, Panels, and Distinguished Lectures

Delivered 35+ invited presentations and keynote-level talks at leading universities, federal agencies, and international conferences.

- Keynote-associated contribution, International Symposium on Automation and Robotics in Construction (ISARC 2024)
- Panelist, ASME/IMECE (Risk and Safety for Human–Robot Collaboration)
- Panelist, NSF National Robotics Initiative PI Meeting
- Federal and national forums: NIOSH, OSHA/NIOSH/A3 Alliance, NORA Construction Sector Council, CDC Symposiums, NOIRS
- Universities (U.S. and international): including Wisconsin, Saint Louis University, Lamar University, George Mason, KAIST, National Tsing Hua University, and others

Editorial and Scholarly Leadership

- Academic Editor, *Journal of Sensors*
- Editorial and leadership roles in major conferences (ASME, IEEE, ACC)
- Longstanding reviewer for leading journals (IEEE, ASME, IFAC) and conferences

Selected Honors and Recognition

1. Best Presentation Award, International Online Conference on Biomimetics (2025)
2. Second-Best Paper Award, International Symposium on Automation and Robotics in Construction (ISARC) (2024)
3. Alice Hamilton Research Award (NIOSH) – Nominated (2024, 2025)
4. IEEE Consumer Electronics Society Chester Sall Award (Second Place Best Paper) (2015)
5. Outstanding Teacher Award, WVU Statler College of Engineering (2016)
6. George W. Weaver Award for Excellence in Teaching, WVU (2015)

Student Success, Teaching, and Workforce Development

Teaching Excellence and Educational Leadership

- Strong record of teaching effectiveness with high student evaluations:
  - National Tsing Hua University: 4.9/5.0
  - West Virginia University: 4.43/5.0
- Recipient of multiple teaching awards:
  - *Outstanding Teacher Award*, WVU Statler College of Engineering (2016)
  - *George W. Weaver Award for Excellence in Teaching* (2015)
  - *Magoon Teaching Assistant Award*, Purdue University (2005)
- Developed and led interdisciplinary, research-integrated educational initiatives aligned with workforce needs in robotics, mechatronics, and cyber-physical systems
- Extensive teaching portfolio across undergraduate and graduate programs, covering robotics, mechatronics, control systems, embedded systems, and engineering fundamentals

Curriculum Development and Innovation

- Developed and delivered modern, interdisciplinary curricula integrating:
  - Robotics, automation, and cyber-physical systems
  - AI-enabled control and intelligent systems
  - Embedded systems and mechatronics

Student Mentorship and Workforce Development

- Mentored Ph.D., M.S., postdoctoral scholars, and research fellows, with placements in:
  - Academia (faculty members in U.S. and international institutions)
  - Federal agencies (NIOSH, NASA, NIH)
  - Industry (robotics, advanced manufacturing, and engineering firms)
- Supervised and advised 600+ undergraduate students, supporting academic success, research engagement, and career development
- Led interdisciplinary research programs and student projects in collaboration with multiple institutions, including work on:
  - Autonomous systems and drone swarming for safety applications
  - Human–robot collaboration and motion prediction
  - Robotics competitions and complex system design (e.g., NASA Sample Return Challenge)

Outreach, Engagement, and Pipeline Development

- Active leadership in STEM outreach and community engagement, including:
  - Judge for national competitions (Real World Design Challenge, WV Science Bowl)
  - Support K–12 robotics programs/competitions and engineering outreach initiatives